The Derivative from First Principles

ANSWER KEY

Section 1: The Limit Definition of the Derivative

1 State the standard limit definition of $f'(x)$.

a. $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ ✓

b. $f'(x) = \lim_{h \to 0} \frac{f(x) - f(h)}{h}$

c. $f'(x) = \frac{f(x+h) - f(x)}{h}$

d. $f'(x) = \lim_{x \to 0} \frac{f(x+h) - f(x)}{x}$

The derivative represents the instantaneous rate of change defined by the limit of the difference quotient as $h \to 0$.

2 Use first principles to find the derivative of $f(x) = 5x^2 - 3x$.

a. $10x - 3$ ✓

b. $10x$

c. $5x - 3$

d. $10x + 3$

$$\lim_{h \to 0} \frac{5(x+h)^2 - 3(x+h) - (5x^2 - 3x)}{h} = \lim_{h \to 0} \frac{10xh + 5h^2 - 3h}{h} = \lim_{h \to 0} (10x + 5h - 3) = 10x - 3.$$

3 Use the alternative limit definition $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3$.

a. 12 ✓

b. 8

c. 6

d. 4

$$\lim_{x \to 2} \frac{x^3 - 8}{x - 2} = \lim_{x \to 2} \frac{(x-2)(x^2 + 2x + 4)}{x - 2} = 2^2 + 2(2) + 4 = 12.$$

Section 2: First Principles for Radicals and Reciprocals

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4 Find $f'(x)$ from first principles for $f(x) = \frac{1}{x}$.

a. $-1/x^2$ ✓

b. $1/x^2$

c. $-1/x$

d. $-2/x^2$

$\lim_{h\to 0} \frac{1/h \cdot (x-(x+h))}{x(x+h)} = \lim_{h\to 0} \frac{-h}{hx(x+h)} = \lim_{h\to 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}$.

5 Find $f'(x)$ from first principles for $f(x) = \sqrt{x}$ $(x > 0)$.

a. $\frac{1}{2\sqrt{x}}$ ✓

b. $\frac{1}{\sqrt{x}}$

c. $-\frac{1}{2\sqrt{x}}$

d. $\frac{2}{\sqrt{x}}$

$\lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h} = \lim_{h\to 0} \frac{(x+h)-x}{h(\sqrt{x+h}+\sqrt{x})} = \frac{1}{2\sqrt{x}}$.

6 Evaluate $\lim_{h\to 0} \frac{(2+h)^5 - 32}{h}$ by recognizing it as a derivative.

a. 80 ✓

b. 40

c. 32

d. 16

This is $f'(2)$ for $f(x) = x^5$. Since $f'(x) = 5x^4$, $f'(2) = 5(2^4) = 5(16) = 80$.

Section 3: Differentiability and Continuity

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7 Which statement correctly describes the relationship between differentiability and continuity?

a. **Differentiability implies continuity, but continuity does not imply differentiability ✓**

b. Continuity implies differentiability

c. They are completely independent properties

d. A function is differentiable if and only if it is continuous

If f is differentiable at c, it must be continuous at c. However, functions like $f(x) = |x|$ are continuous at 0 but not differentiable there.

8 Why does $f(x) = |x - 3|$ fail to be differentiable at $x = 3$?

a. **The left-hand and right-hand derivative limits are unequal ($-1 \neq 1$) ✓**

b. $f(3)$ is undefined

c. f is discontinuous at $x = 3$

d. The tangent line is horizontal

Left derivative is $\lim_{h \to 0^-} \frac{-(3+h-3)}{h} = -1$. Right derivative is $+1$. Since they differ, a sharp corner exists and $f'(3)$ does not exist.

9 At $x = 0$, the function $f(x) = x^{1/3}$ has which type of non-differentiability?

a. **Vertical tangent ($f'(x) \to \infty$) ✓**

b. Corner / sharp turn

c. Cusp

d. Removable discontinuity

$f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}}$. As $x \to 0$, $f'(x) \to +\infty$. The tangent line is vertical, so f is not differentiable.

Section 4: Geometric and Physical Interpretations

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10 What geometric feature does the derivative $f'(a)$ represent?

a. **The slope of the tangent line to $y = f(x)$ at $x = a$** ✓

b. The area under the curve from 0 to a

c. The secant line slope between 0 and a

d. The curvature at $(a, f(a))$

The derivative $f'(a)$ is geometrically defined as the exact slope of the tangent line to the curve $y = f(x)$ at $(a, f(a))$.

11 A particle moves along a line with position $s(t) = 4t^2 - 3t + 1$. What is its instantaneous velocity at $t = 2$?

a. 13 ✓

b. 11

c. 16

d. 10

$v(t) = s'(t) = 8t - 3$. At $t = 2$: $v(2) = 8(2) - 3 = 13$.

12 Find the equation of the tangent line to $y = x^2 - 4x$ at the point where $x = 3$.

a. $y = 2x - 9$ ✓

b. $y = 2x - 3$

c. $y = -x$

d. $y = 4x - 15$

$y(3) = 9 - 12 = -3$. $y' = 2x - 4 \Rightarrow y'(3) = 2$. Line: $y - (-3) = 2(x - 3) \Rightarrow y = 2x - 9$.

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